Unisoc Confidential For longcheer

Android 11.0 YLog Capture Guide

Issue V1.3

Date 2022-07-26

UNISOC (Shanghai) Technologies Co., Ltd.

Copyright © UNISOC (Shanghai) Technologies Co., Ltd. All rights reserved.

All data and information contained in or disclosed by this document is confidential and proprietary information of UNISOC (Shanghai) Technologies Co., Ltd. (hereafter referred as UNISOC) and all rights therein are expressly reserved. This document is provided for reference purpose, no license (express or implied, by estoppel or otherwise) to any intellectual property rights is granted by this document, and no express and implied warranties, including but without limitation, the implied warranties of fitness for any particular purpose, and non-infringement, as well as any performance. By accepting this material, the recipient agrees that the material and the information contained therein is to be held in confidence and in trust and will not be used, copied, reproduced in whole or in part, nor its contents revealed in any manner to others without the express written permission of UNISOC. UNISOC may make any changes at any time without prior notice. Although every reasonable effort is made to present current and accurate information, UNISOC makes no guarantees of any kind with respect to the matters addressed in this document. In no event shall UNISOC be responsible or liable, directly or indirectly, for any damage or loss caused or alleged to be caused by or in connection with the use of or reliance on any such content.

Please refer to the UNISOC Documents in the UNISOC Deliverables for the use of the Deliverables. Any loss caused by the modification, customization or use of the UNISOC Deliverables in violation of the instructions in the UNISOC Documents shall be undertaken by those who conduct so. The performance indicators, test results and parameters in the UNISOC Deliverables are all obtained in the internal development and test system of UNISOC and are only for the reference. Before using UNISOC Deliverables commercially or conducting mass production of the Deliverables, comprehensive testing and debugging in combination with its own software and hardware test environment are pre-requisite.

Trademarks

UNISOC, , Spreadtrum, SPRD, RDA, and other UNISOC trademarks are trademarks of UNISOC (Shanghai) Technologies Co., Ltd. and/or its affiliated companies.

The Bluetooth® word mark and logos are registered trademarks owned by Bluetooth SIG, Inc. and any use of such marks by UNISOC (Shanghai) Technologies Co., Ltd. is under license.

All other trademarks and trade names mentioned in this document are the property of their respective owners.

Unisoc Confidential For longcheer

UNISOC (Shanghai) Technologies Co., Ltd.

About this document

Purpose

This document describes the usage of YLog on Android 11.0.

Intended audience

This document is intended for users who capture logs with YLog on Android 11.0.

Symbol conventions

The symbols that may be found in this guide are defined in the following table.

Symbol	Description
NOTE	Calls attention to important information, best practices and tips. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.
CAUTION	Calls attention to error-prone operations. CAUTION is used to address information not related to personal injury, equipment damage, and environment deterioration.
WARNING	Calls attention to irreversible operations. WARNING is used to address information not related to personal injury and environment deterioration.

Unisoc Confidential For longcheer

Acronyms and abbreviations

Acronym and Abbreviation	Full Name
AGDSP	Audio DSP
AP	Application Processor
CP	Communication Processor
DUT	Device Under Test
WCN	Wireless Connectivity Network

 Copyright © UNISOC (Shanghai) Technologies Co., Ltd.

Change History

Issue	Date	Description
V1.3	2022-7-26	Add UMS9620 related information, mainly modifying **4.1.1.6 Custom scene**, **4.1.2.1 Debug setting & tools**, **4.4.2 Export log with SD card**.
V1.2	2021-03-10	• Align the content with that in the Chinese version. • Rename the document from *Android11 YLog4.1 Capture User Guide* to *Android 11.0 YLog Capture Guide*.
V1.1	2020-08-04	Alignment, grammar, spelling and sentence correction.
V1.0	2020-07-15	Figure modified, Content modified.

Keywords

YLog, Engineering mode, YLog export, Sysdump.

Unisoc Confidential For longcheer

Contents

1 YLog introduction 1
1.1 Overview 1
1.2 Log types 2
2 YLog release notes 3
2.2 YLog version history 3
2.3 Version mapping between Android and YLog 4
3 Environment configuration 6
3.1 Test resource preparation 6
3.2 PC configuration 6
3.3 DUT configuration 6
4 Instructions to YLog 7
4.1 YLog settings 7
4.1.1 Scene settings 7
4.1.2 Advanced settings 13
4.2 Log capture 15
4.2.1 Capture log in UserDebug version 15
4.2.2 Capture log in User version 15
4.3 Log storage 15
4.4 Log export 16
4.4.1 Export log with tool 16
4.4.2 Export log with SD card 17
4.5 Log replay 19
4.5.1 AP log replay 20
4.5.2 Modem log replay 20
4.5.3 Connectivity log replay 21
5 Log capture in special scene 22
5.1 Sysdump log for frozen screen or system crash 22
5.1.1 YLog settings 22
5.1.2 Log capture 23
5.1.3 Log storage 25
5.2 Memory log of assert problem 26
5.2.1 YLog settings 26
5.2.2 Log capture 26
5.2.3 Log storage 27
5.3 Audio log 28
5.3.1 YLog settings 28
5.3.2 Log capture 28

Unisoc Confidential For longcheer

5.3.3 Log storage 28
5.3.4 Log replay 28
5.4 Power consumption problem log 31
5.4.1 YLog settings 31
5.4.2 Log export 31
5.5 MonkeyTest log 31
5.5.1 YLog settings 31
5.5.2 Log export 31
6 Points for attention 32
7 Frequently asked questions 33

Unisoc Confidential For longcheer

List of figures

Figure 1-1 Entering YLog 1
Figure 2-1 YLog version number 3
Figure 4-1 Settings 7
Figure 4-2 Log tool description 8
Figure 4-3 Audio scene on chip with independent AGDSP 10
Figure 4-4 Audio scene on chip without independent AGDSP 10
Figure 4-5 User scene 11
Figure 4-6 Custom scene for chips with independent AG-DSP 12
Figure 4-7 Custom scene for chips without independent AG-DSP 12
Figure 4-8 Debug settings & tools 13
Figure 4-9 User version settings 14
Figure 4-10 User version log settings 14
Figure 4-11 ADB command to view YLog directory 15
Figure 4-12 YLog directory for chips with independent AGDSP 15
Figure 4-13 YLog directory on chips without independent AGDSP 16
Figure 4-14 Export tool folder 17
Figure 4-15 Command window 17
Figure 4-16 Log storage location (UMS9620) 18
Figure 4-17 Debug settings & tools 18
Figure 4-18 Log auto copy to SD card setting 19
Figure 4-19 YLog directory 20
Figure 4-20 APLog replay 20
Figure 4-21 Modem log replay 20
Figure 5-1 Reset setting options 23
Figure 5-2 Sydump screen with sufficient SD space 24
Figure 5-3 Sysdump screen with no SD card inserted 24
Figure 5-4 Auto Identify AP Dump Pt window 25
Figure 5-5 Dump to PC window 25

Unisoc Confidential For longcheer

Figure 5-6 Sysdump log list 26
Figure 5-7 Debug setting & tools 27
Figure 5-8 Assert mem log 28
Figure 5-9 Audio mem log 29
Figure 5-10 Audio dsp transfer 30
Figure 5-11 DSP PCM data 30
Figure 5-12 ARM PCM data 31

Unisoc Confidential For longcheer

List of tables

Table 1-1 Log type 2
Table 2-1 YLog version history 3
Table 2-2 Version mapping between Android and YLog 5
Table 4-1 Logs captured in each scene on chip with independent AGDSP 8
Table 4-2 Logs captured in each scene on chip without independent AGDSP 9

Unisoc Confidential For longcheer

1 YLog introduction

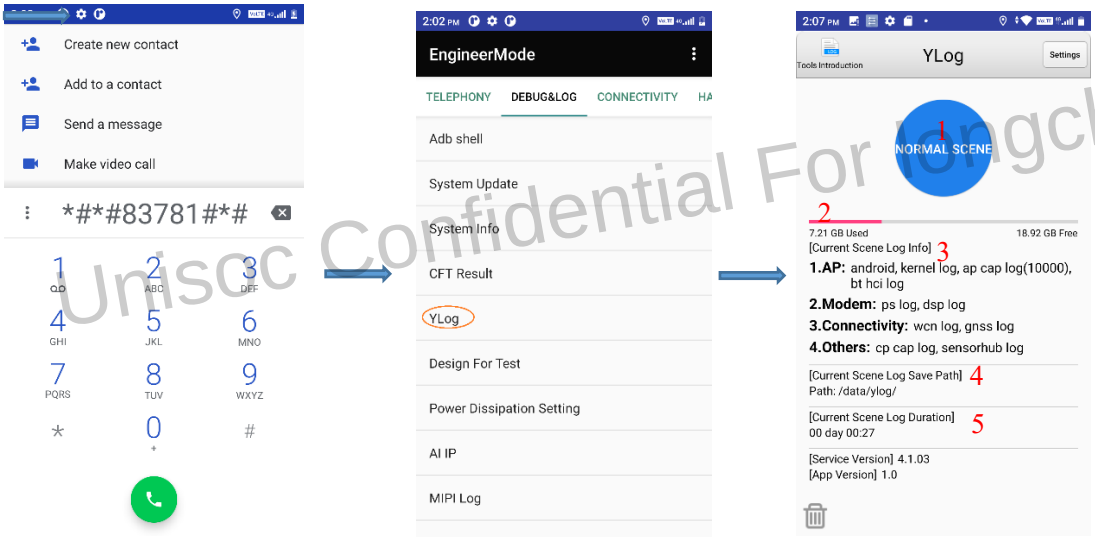
1.1 Overview

There are two log capture modes on UNISOC platform, namely online mode and offline mode. In the online mode, you can capture logs through PC. In the offline mode, you can work offline to get logs, and the tool was previously called Slog and later upgraded to YLog.

YLog is a daemon written in C language, which runs on the native layer of Android system to collect various logs output by each module in devices. These logs are stored in device memory or SD card, which is convenient for developers to view and analyze issues.

Type ***#*#83781#*#*** on the dialer to enter Engineer Mode, and then tap **DEBUG&LOG** > **YLog** to enter the main screen of YLog, as shown in **Figure 1-1**.

Figure 1-1 Entering YLog



Unisoc Confidential For Longcheer

1. **YLog Switch and Current Scene**: blue means on, while gray off.
2. **Internal Storage Usage:** red indicates the used space, while gray the remaining space.
3. **Current Scene Log Info:** it displays the modules to output logs in current scene, such as AP log, Modem log, which is related with the selected scene.
4. **Current Scene Log Save Path:** it indicates the current log storage location.
5. **Current Scene Log Duration:** it indicates the duration of log capture. The timer is cleared once log capture stops.

1.2 Log types

There are two categories for UNISOC chips, namely chips with an independent AGDSP module and chips without an independent AGDSP module.

- Chips with an independent AGDSP module have separate Audio DSPs, for example, UMS512.
- Chips without an independent AGDSP module use one DSP for Audio and PHY_GSM, for example, SC9863A, SC9832E, and SC7731E.

For chips with an independent AGDSP module, the types of logs captured by YLog are as shown in **Table 1-1**.

Table 1-1 Log type

Log Category	Log Name	Log Description
AP	android log	Logs in the five buffer: main, system, radio, events and crash
	kernel log	Logs of reading /proc/kmsg
	ap cap log	Tcpdump log on the AP side
	bt hci log	**Bluetooth**® log generated when a host interacts with a controller
Modem	ps log	Protocol stack log
	arm pcm log	Voice call related arm pcm data
	agdsp log	Log of the Audio DSP module
	agdsp pcm log	pcm data of the Audio DSP module
	dsp log	Physical layer log
Connectivity	wcn log	Include logs related with Bluetooth, Wi-Fi, and FM
	gnss log	Positioning related log
Others	cp cap log	Tcpdump log on CP side
	sensorhub log	Sensor hub processor log
	dsp pcm log	Voice call related dsp mem data

Unisoc Confidential For longcheer

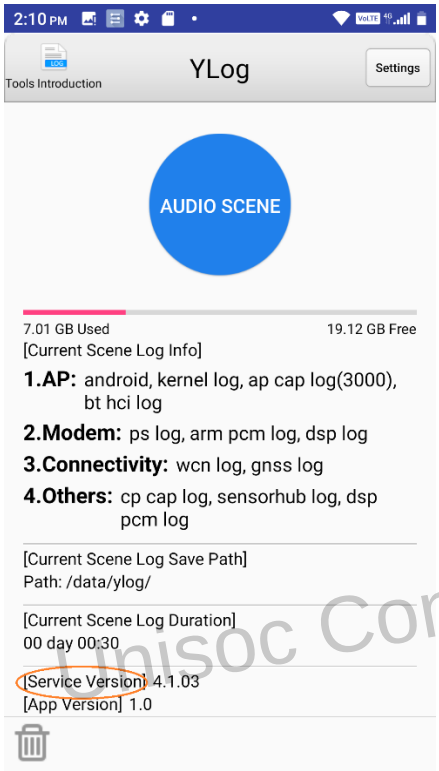
For chips without an independent AGDSP module, there is no agdsp log and agdsp pcm log in the logs captured by YLog, and the remaining is the same as that in **Table 1-1**.

 Copyright © UNISOC (Shanghai) Technologies Co., Ltd.

2 YLog release notes

The version number of YLog is available on the YLog main screen, as shown in **Figure 2-1**.

Figure 2-1 YLog version number



Unisoc Confidential For longcheer

2.2 YLog version history

YLog version history is shown in **Table 2-1**.

Table 2-1 YLog version history

YLog Version	Revisions History
1.0	The initial release.
2.0	• Solved the log loss problem of YLog. • Added the log current-limiting function. • Added the log compression function to reduce I/O throughput and disk occupation. • Added the function of proactively capturing and protecting special logs, such as Monkey first error log, file handle leakage log.

YLog Version	Revisions History
2.1	• Modified AP log storage location: AP log is now stored under the directory **ylog/ap**, while it was stored under the **ylog** directory in previous versions. • Modified the file name of the AP Log directory: once the AP Log reaches the size limit of a single directory, a new log directory is created. The name of the new log directory includes log start and end time. • Optimized AP Log quota management strategy: there is no storage ratio limit for various logs in case of enough space, whereas default limit is restored in case of space constrains. • The newly added AP poweron log is saved separately in the **ylog/ap/poweron** directory.
3.0	• Expanded log compression scope to improve IO writing efficiency and reduce disk consumption as well. Previously only android log compression was supported, and now compression of kernel log, sysinfo log, hcidump log, and tcpdump log is also supported. • All types of logs are compressed into the same file, which reduces system overhead and facilitates log management. • Optimized architecture which supports configurable log source loading and is easy to add new log type output. • Cancelled the renaming operation of history log directory to solve the problem of history log loss. • Optimized the communication instructions with Log manger to reduce the probability of communication problems. • Modified the YLog storage strategy by removing the limit of maximum 40% space occupation. • Optimized tcpdump and hcidump log mechanism to avoid format error. • Optimized the log capture mechanism in the factory mode to ensure that the factory log can be captured.
4.0	• Added the function of capturing uboot log and lastlog (the android log and kernel log at last restart). • Fixed the issue of log loss caused by Android or YLog crash. • Optimized Log on / off/ clear performance.
4.1	• Changed the default log storage location to **data/ylog**, and deleted SD card as a log storage location. • Deleted the Get all log function. • Added the function of Log Copy to SD Card.

Unisoc Confidential For longcheer

2.3 Version mapping between Android and YLog

Version mapping between Android and YLog is given in **Table 2-2**.

Table 2-2 Version mapping between Android and YLog

Android version	Branch	YLog version
Android 4.4	sprdroid4.4	SLog
Android 6.0	sprdroid6.0	SLog & YLog
Android 7.0	Sprdroid7.0	YLog 2.0 & YLog 2.1
Android 8.1	sprdroid8.1_trunk_18a	YLog 2.1
	sprdroid8.1_trunk_18b	YLog 2.1
Android 9.0	sprdroid9.0_trunk_18c	YLog 2.1
	sprdroid9.0_trunk_19a	YLog 3.0
	sprdroid9.0_trunk_19b	YLog 4.0
Android 10.0	sprdroidq_trunk	YLog 4.0
Android 11.0	sprdroidr_trunk	YLog 4.1

Unisoc Confidential For longcheer

3 Environment configuration

3.1 Test resource preparation

A Device Under Test (DUT), a test version, a USB cable, and a PC.

3.2 PC configuration

- Install the ADB environment and driver for PC to identify the port after inserting the USB cable in the DUT.
- Download Log4Android2PC.
- Install python.

3.3 DUT configuration

- Download the test version to the DUT and boot the DUT normally.
- Insert the SD card into the DUT in special test scenarios generating a large amount of logs, such as Monkey test.

Unisoc Confidential For longcheer

4 Instructions to YLog

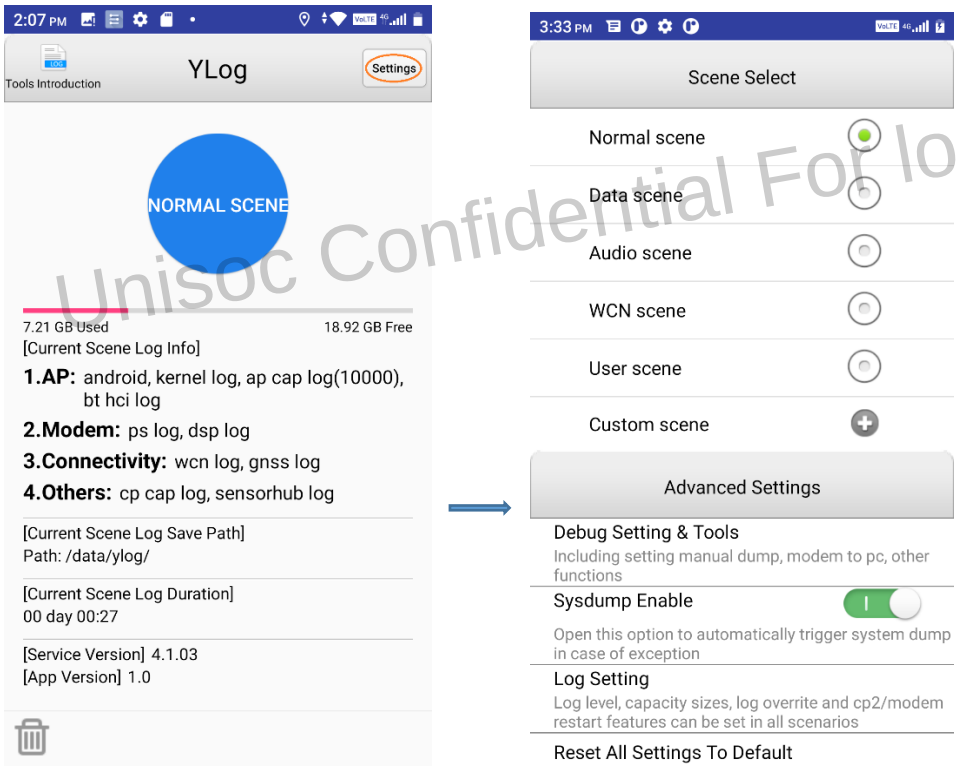
There are five stages in using YLog, namely YLog settings, log capture, log storage, log export, and log replay. This section describes the log capture instructions in general scenes using YLog, whereas **5 Log capture in special scene** describes the log capture instructions in special scenes using YLog.

There are UserDebug and User versions for UNISOC software. The UserDebug version is used in debugging phase, whereas the User version is used in shipping and mass production phases.

4.1 YLog settings

Click the **Settings** button in the upper-right corner of the YLog main screen to set YLog. There are **Scene Select** and **Advanced Settings** in YLog settings, as shown in **Figure 4-1**.

Figure 4-1 Settings

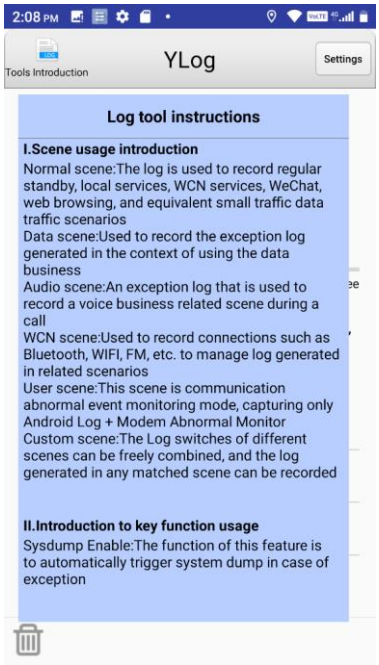


Unisoc Confidential For longcheer

4.1.1 Scene settings

For scene introduction, see **Tools instruction** in the upper-left corner of YLog main screen, as shown in **Figure 4-2**.

Figure 4-2 Log tool description



- For chips with an independent AGDSP, the logs captured in each scene are as shown in **Table 4-1**.

Table 4-1 Logs captured in each scene on chip with independent AGDSP

Log Category	Log Name	Normal Scene	Data Scene	Audio Scene	WCN Scene	User Scene
AP	android log	√	√	√	√	√
	kernel log	√	√	√	√	√
	ap cap log	√	√	√	√	√
	bt hci log	√	√	√	√	√
Modem	ps log	√	√	√	√	×
	arm pcm log	×	×	√	×	×
	agdsp log	√	×	√	×	×
	agdsp pcm log	√	×	√	×	×
	dsp log	√	√	√	√	×
Connectivity	wcn log	√	√	√	√	×
	gnss log	√	√	√	√	×
Others	cp cap log	√	√	√	√	×
	sensorhub log	√	√	√	√	×

Unisoc Confidential For longcheer

Log Category	Log Name	Normal Scene	Data Scene	Audio Scene	WCN Scene	User Scene
	dsp pcm log	×	×	√	×	×
	Modem event monitor	×	×	×	×	√

- For chips without an independent AGDSP, the logs captured in each scene are as shown in **Table 4-2**.

Table 4-2 Logs captured in each scene on chip without independent AGDSP

Log Category	Log Name	Normal Scene	Data Scene	Audio Scene	WCN Scene	User Scene
AP	android log	√	√	√	√	√
	kernel log	√	√	√	√	√
	ap cap log	√	√	√	√	√
	bt hci log	√	√	√	√	√
Modem	ps log	√	√	√	√	×
	arm pcm log	×	×	√	×	×
	dsp log	√	√	√	√	×
Connectivity	wcn log	√	√	√	√	×
	gnss log	√	√	√	√	×
Others	cp cap log	√	√	√	√	×
	sensorhub log	√	√	√	√	×
	dsp pcm log	×	×	√	×	×
	Modem event monitor	×	×	×	×	√

Unisoc Confidential For longcheer

4.1.1.1 Normal scene

This is the default scene of YLog, which is used to capture the logs generated in scenarios consuming comparably less traffic data, such as standby, local business, WCN service, wechat, web browsing.

4.1.1.2 Data scene

It is used to record the abnormal log generated in the scenarios of using data services.

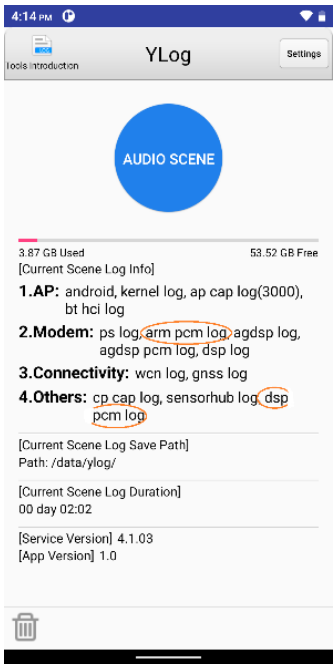
- On chips with an independent AGDSP, data scene differs from normal scene in that data scene does not output AGDSP log.
- On chips without an independent AGDSP, data scene and normal scene outputs the same log types.

4.1.1.3 Audio scene

It is used to record the abnormal log generated in the voice service related scenarios during the call.

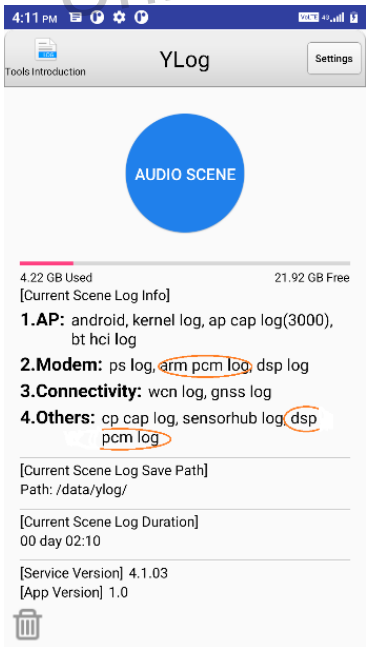
Compared with normal scene, audio scene additionally outputs arm pcm log, dsp pcm log, as shown in **Figure 4-3** and **Figure 4-4**.

Figure 4-3 Audio scene on chip with independent AGDSP



Unisoc Confidential For longcheer

Figure 4-4 Audio scene on chip without independent AGDSP



4.1.1.4 WCN scene

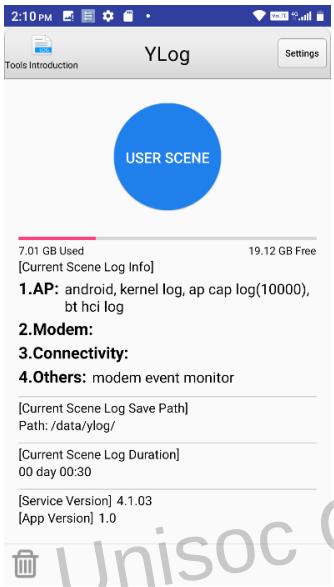
It is used to record logs generated in connection management scenarios such as Bluetooth, Wi-Fi, and FM.

- For chips with an independent AGDSP, WCN scene differs from normal scene in that WCN scene does not output AGDSP log.
- For chips without an independent AGDSP, WCN scene and normal scene output the same log types.

4.1.1.5 User scene

This scene is abnormal communication event monitoring mode, which only captures Android Log and Modem Abnormal Monitor, as shown in **Figure 4-5**.

Figure 4-5 User scene



Unisoc Confidential For longcheer

4.1.1.6 Custom scene

Combine the log switches under different scenes freely to record the logs generated in any scene.

You can rename the custom scene in the input box, and then enter the editing screen, as shown in **Figure 4-6**, **Figure 4-7**.

Figure 4-6 Custom scene for chips with independent AG-DSP

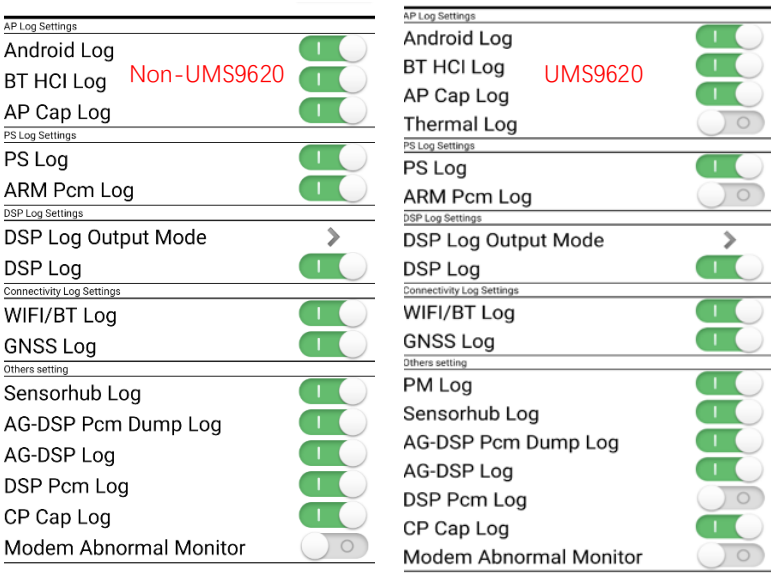
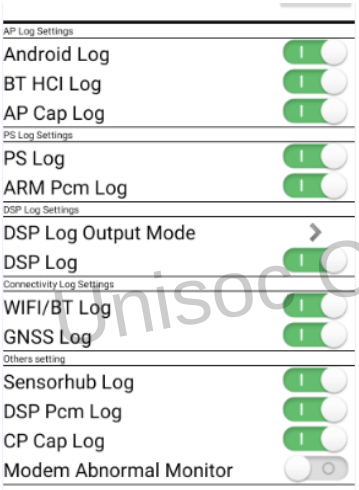


Figure 4-7 Custom scene for chips without independent AG-DSP



Unisoc Confidential For longcheer

- **AP Log Settings:** the logs on the AP side include Android Log, BT HCI Log, and AP Cap Log. The Android Log switch can be turned on to generate main log, system log, radio log, event log, kernel log, and crash log.
- **PS Log Settings:** it is used to turn on or turn off PS Log and ARM Pcm Log.

  Turning on ARM Pcm Log can achieve simultaneous capture of WCN Log and Voice Log. The PS Log switch needs to be turned on while capturing ARM Pcm Log.
- **DSP Log Settings:** it is used to turn on or turn off DSP Log and set DSP Log output mode. You can select between Output from UART or Output with Modem log.
- **Connectivity Log Settings:** it is used to tur on or turn off WIFI/BT Log and GNSS Log.
- **Others Setting:** it is used to turn on or turn off Sensorhub Log, DSP Pcm Log and Cap Log. With DSP Pcm Log turned on, you can capture WCN Log and audio Log at the same time. The DSP Log needs to be turned on when you capture DSP Pcm log. Capture DSP Pcm Log to analyze call quality related issues.
- **Modem abnormal monitor:** it is used to turn on or turn off the function of Modem Abnormality Monitoring.

PS Log and DSP Log are not compatible with Modem Abnormal Monitor.PS Log and DSP Log is turned off automatically if Modem Abnormal Monitor is turned on, and Modem Abnormal Monitor is turned off if PS Log and DSP Log are turned on. Enabling Modem Abnormal Monitor, which does not save Modem Log or capture log through PC, will enable the function of communication abnormal event.

NOTE

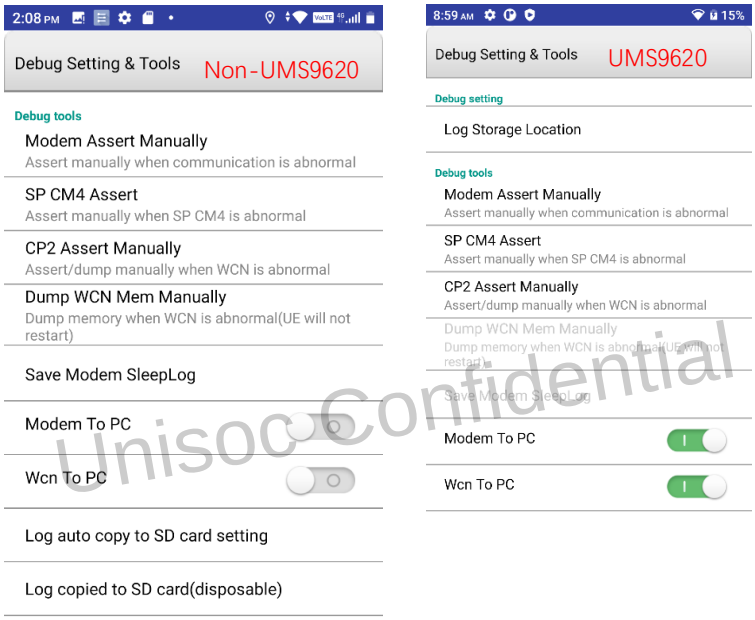
The Log switch options of the previous scene is turned on by default when changing from other scene to custom scene.

4.1.2 Advanced settings

4.1.2.1 Debug setting & tools

By default, Modem and WCN log is captured offline. To capture Modem and WCN log online, enable **Modem to PC** and **Wcn to PC** in **YLog > Settings > Debug Settings & Tools**, as shown in **Figure 4-8**.

Figure 4-8 Debug settings & tools



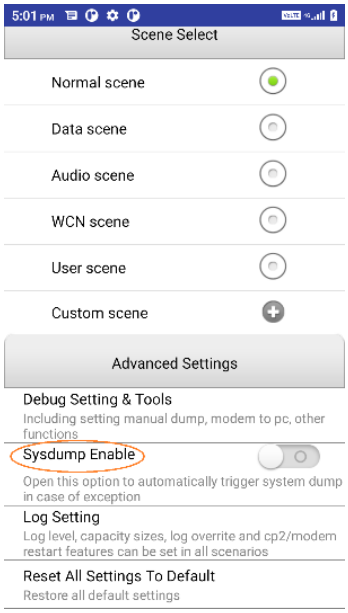
Unisoc Confidential For longcheer

4.1.2.2 Sysdump enable

Sysdump Enable is used to enable the Sysdump function to output Sysdump log.

- In UserDebug version, Sysdump Enable is enabled by default.
- In User version, Sysdump Enable is disabled by default. To output Sysdump log, you need to enable it in the Settings menu before testing, as shown in **Figure 4-9**.

Figure 4-9 User version settings

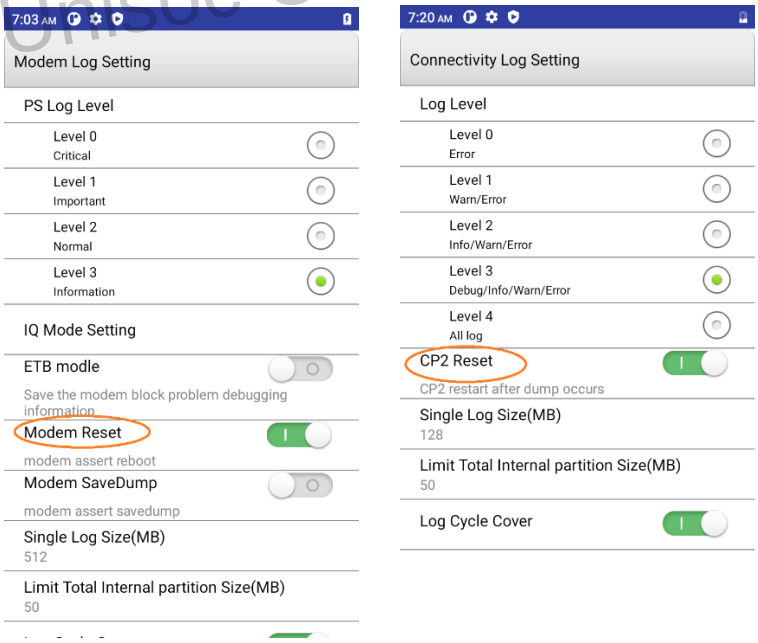


4.1.2.3 Log setting

- In UserDebug version, **Modem Reset** and **CP2 Reset** are turned off by default.
- In User version, **Modem Reset** and **CP2 Reset** are turned on by default, which needs to be turned off before testing in **YLog** > **Settings** > **Log Setting** > **Modem Log Setting /Connectivity Log Setting** as shown in **Figure 4-10**.

Unisoc Confidential For Longscheer

Figure 4-10 User version log settings



NOTE

The log setting screens vary slightly for different projects. For details, refer to the actual UI.

4.2 Log capture

4.2.1 Capture log in UserDebug version

By default, YLog switch is blue and turned on in UserDebug version. You only need to confirm that the log size increases gradually along with the increase in log duration to make sure YLog works properly.

4.2.2 Capture log in User version

In User version, turn on the YLog switch manually, because it is gray and turned off by default. Make sure that the log size gradually increases along with the increase in log duration.

4.3 Log storage

The YLog folder is generated under the Data directory, which can be viewed through ADB command, as shown in **Figure 4-11**.

Figure 4-11 ADB command to view YLog directory

```
C:\Users\[redacted]>adb root
adbd is already running as root

C:\Users\[redacted]>adb shell
sp9832e_1h10_go:/ # cd data
cd data
sp9832e_1h10_go:/data # ls
ls
adb            corefile      nfc                        system_ce
anr            dalvik-cache  ota                        system_de
apex           data          ota_package                tombstones
app            drm           preloads                   unencrypted
app-asec       gsi           property                   user
app-ephemeral  local         resource-cache             user_de
app-lib        lost+found    rollback                   vendor
app-private    media         rollback-observer          vendor_ce
app-staging    mediadrm      server_configurable_flags  vendor_de
backup         misc          sprdinfo                   ylog
bootchart      misc_ce       ss
cache          misc_de       system
sp9832e_1h10_go:/data # cd ylog
cd ylog
sp9832e_1h10_go:/data/ylog # ls
```

ngcheer

- For chips with an independent AGDSP module, the files under the YLog directory are as shown in **Figure 4-12**.

  Figure 4-12 YLog directory for chips with independent AGDSP

```
ums512_1h10:/data/ylog $ ls
ls
ap      connectivity  loguid  phone.info  rate_statistic.txt  sensorhub
audio   journal.log   modem   poweron     screen.jpg          ylog.conf
```

- For chips without an independent AGDSP module, the files in the YLog directory are as shown in **Figure 4-13**.

Figure 4-13 YLog directory on chips without independent AGDSP

```
s9863a3h10:/data/ylog # ls
ls
ap            loguid      poweron             sensorhub
connectivity  modem       rate_statistic.txt  ylog.conf
journal.log   phone.info  screen.jpg
```

- **AP:** the folder to store AP logs.
- **Audio:** the folder to store AGDSP related logs.
- **Connectivity:** the folder to store the wcn log in corresponding scene, including .log and .dmp files.
- **Modem:** the foler to store the modem log in corresponding scene, including .log, .bin, and .dmp files.
- **Poweron:** the folder to store the modem log within 2 minutes after boot in the last 3 times.
- **Sensorhub:** the folder to store the sensor hub log in corresponding scene.

4.4 Log export

4.4.1 Export log with tool

In UserDebug version, the USB debugging switch is turned on by default. However, in User version, it is turned off by default, so you need to turn it on manually. There are two ways to turn on the USB debugging switch:

- Go to **Settings** > **About Phone** > **Build number**, click **Build number** for 7 times, go back to **Settings > System > Developer Options**, and then turn on the **USB debugging** switch.
- Go to **EngineerMode** > **DEBUG&LOG**, and turn on **USB Debug**.

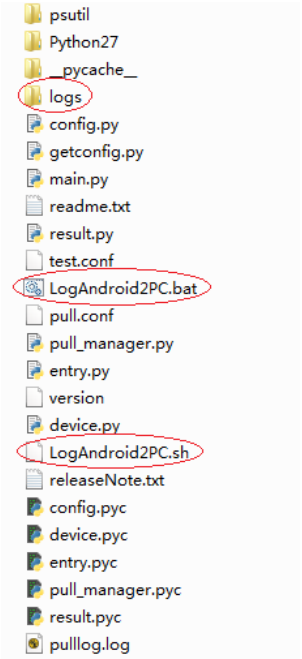
Unisoc Confidential For longcheer

The steps to export YLog with Log4Android2PC are as follows.

Step 1 Insert a USB cable into the DUT to connect to PC. The DUT prompts **Connected to USB debugging**, and the PC can correctly identify the ADB port.

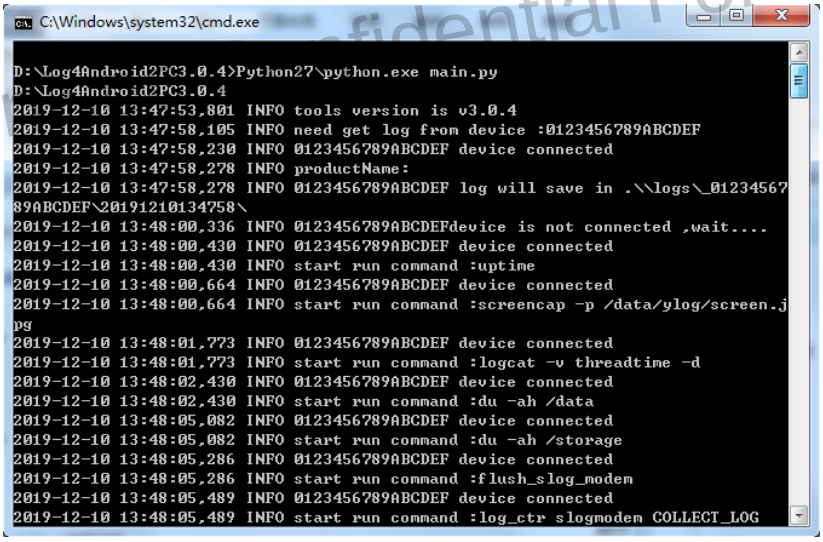
Step 2 Open the Log4Android2PC folder, run **LogAndroid2PC.bat** on Windows system, or **LogAndroid2PC.sh** on Linux system, as shown in **Figure 4-14**.

Figure 4-14 Export tool folder



Log export starts when it pops up a command line window, as shown in **Figure 4-15**. The command line window is closed automatically after log export ends, and logs are stored in the logs folder.

Figure 4-15 Command window



Unconfidential For longcheer

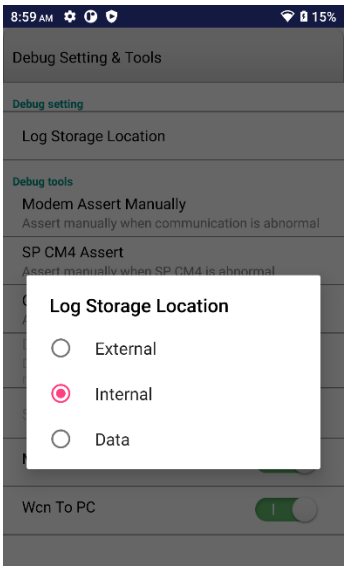
----End

4.4.2 Export log with SD card

If the device supports SD card and the test runs for a long time:

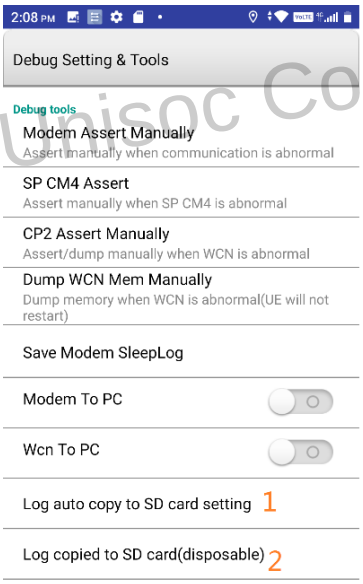
- For UMS9620, you can set **Log Storage Location** to **External**, as shown in **Figure 4-16**, and then copy the log to PC.

  Figure 4-16 Log storage location (UMS9620)



- For non-UMS9620 platforms, you can export logs by copying log files to the SD card and then to the PC, as shown in **Figure 4-17**.

  Figure 4-17 Debug settings & tools



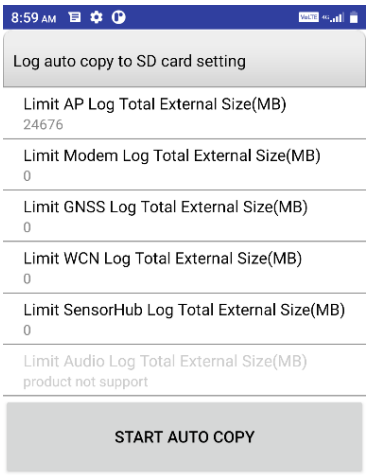
Unisoc Confidential For longcheer

1. **Log auto copy to SD card setting:** once the function is enabled, logs are copied to SD card automatically. When logs are successfully copied to SD card, the corresponding log files under the **/data/ylog** directory are deleted and stored to **/storage/sdcard0/Documents/ylog**.
2. **Log copied to SD card (disposable):** Copy logs to SD card in a single time, which is similar to Auto Log Copy.

Log auto copy to SD card setting

Step 1 Click **Log auto copy to SD card setting**, the screen is as shown in **Figure 4-18**.

Figure 4-18 Log auto copy to SD card setting



Step 2 Allocate the storage size of the remaining space on SD card reasonably before enabling this function.

Only copy the logs under the folders of AP, Modem, Connectivity, Sensorhub and Audio, but not the logs under other folders.

Step 3 After setting, click **START AUTO COPY** at the bottom of the screen.

It pops up a notification of **Log Copy Service is Running** on the status bar. If the notification is not there, it indicates the copy process ends.

Unisoc Confidential For longcheer

----End

Points for attention

- It judges the storage size of SD card and total external size before copying. In case the SD card is full or it reaches the total external size of a specific log:
  - If it supports log overwriting, delete the oldest logs before copying.
  - If it does not support log overwriting, stop copying logs.
- Auto Log Copy ends once the DUT restarts.
- The log files being written is not copied.
- It is not suggested to turn on this function in common test where logs are small and no need to be stored for a long time. In such situation, logs can be saved to the **/data** partition directly.
- This function is not real-time synchronization. You need to view logs in both the **/data/ylog** directory and the **/storage/sdcard0/Documents/ylog** directory when analyzing logs.
- This function is an operation of reading I/O in the background, which may affect the overall device performance. If the device is stuck or shows ANR after enabling this function, close it and test again.

4.5 Log replay

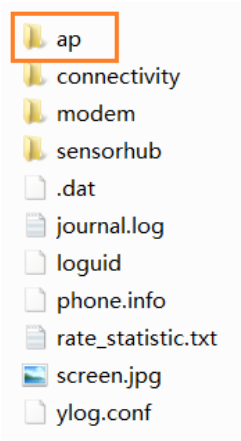
The YLog replay includes the replay of AP Log, Modem Log, and Connectivity Log.

4.5.1 AP log replay

The steps to play back AP logs are as follows:

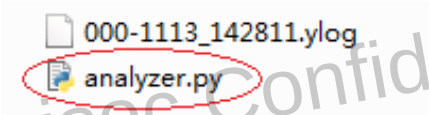
Step 1 Select the AP Log subdirectory under the YLog folder, as shown in **Figure 4-19**.

Figure 4-19 YLog directory



Step 2 Run analyzer.py, and corresponding main.log, radio.log, system.log and other logs are generated in the current directory, as shown in **Figure 4-20**.

Figure 4-20 APLog replay



Unisoc Confidential For longcheer

Step 3 Use Notepad ++ or UltraEdit to open and view AP Log.

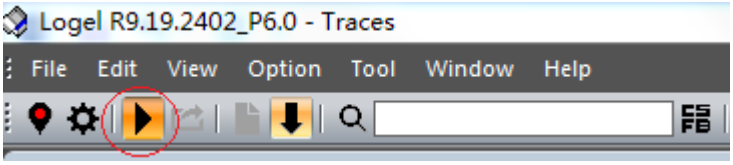
----End

4.5.2 Modem log replay

Step 1 Open Logel and click the open log button in the upper left corner, as shown in **Figure 4-21**.

Step 2 Select .log files in the Modem Log subdirectory in the YLog folder to replay the modem logs exported from YLog.

Figure 4-21 Modem log replay



----End

 Copyright © UNISOC (Shanghai) Technologies Co., Ltd.

4.5.3 Connectivity log replay

Step 1 Open Logel and click the open log button in the upper-left corner, as shown in **Figure 4-21**.

Step 2 Select .log file in the Connectivity Log subdirectory in the YLog folder to play back the connectivity log exported from YLog.

----End

Unisoc Confidential For longcheer

5 Log capture in special scene

This section describes the log capture in special scenes. For YLog capture for general scenes, see **4 Instructions to YLog**.

This chapter mainly describes the log capture differences between special scene and general scene.

5.1 Sysdump log for frozen screen or system crash

The issues of frozen screens are comparably complex, which requires on-site information for issue analysis. Perform the following steps to confirm on-site information:

Step 1 Record the time point and operation steps of the frozen screen, and also take pictures of the frozen screen.

Step 2 Confirm if ADB can be connected by entering **adb devices** in the cmd command line to see if it outputs the DUT information.

Step 3 Use a helper device to call the test device if a SIM card is inserted and check if there is ringtone to estimate if the modem works properly.

Step 4 Press the physical button such as the power button and the volume buttons 2 to 3 times within 5 minutes to see if the screen changes.

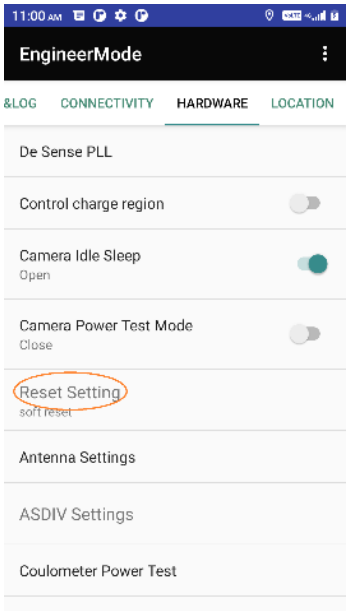
----End

Unisoc Confidential For longcheer

5.1.1 YLog settings

- The prerequisite to trigger sysdump is that the **sysdump enable** switch is turned on. Related settings are shown in **Figure 4-9**.
- If you trigger sysdump using 7S long-press (press and hold the Power button and the volume up button for 7 seconds), you need to select **soft reset** in **Reset Setting**, which can be set in **HARDWARETEST** > **Reset Setting** in Engineer Mode, as shown in **Figure 5-1**.

Figure 5-1 Reset setting options



5.1.2 Log capture

Step 1 Trigger sysdump by below mentioned methods:

- 7S long press: press and hold the Power button and the volume up button for 7 seconds.
- Combination key: press volume up and volume down buttons at the same time, and press the power button 2 times in succession.

Step 2 Sysdump

- In case of enough space in SD:

  It takes several minutes to complete the Dump process after triggering sysdump. When the message **Writing done. Press any key (Exp power key) to continue...** is displayed on the screen, press any button except the power button to restart the DUT as shown in **Figure 5-2**.

Unisoc Confidential For longcheer

Figure 5-2 Sydump screen with sufficient SD space

- In case of insufficient space in SD card:

  When sysdump is triggered in case of insufficient space in SD it prompts **press volume down to format SD Card**. Press the volume down button to format the SD card. Once there is enough space, dump log to SD card.

- In case of no SD card:

  When sysdump is triggered in case of no SD card it prompts **sysdump to sdcard fail, we try to do dump to pc now**, as shown in **Figure 5-3**. In such case, you cannot save the dump file to the SD card, but can export sysdump log through PC.

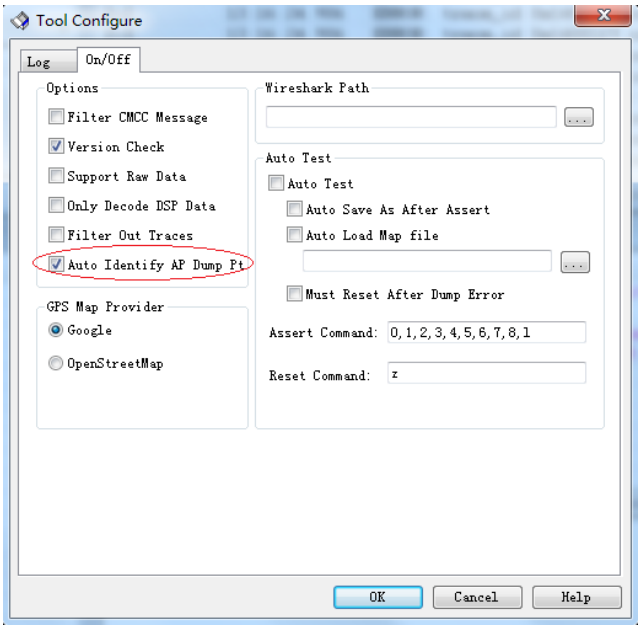
Unisoc Confidential For longcheer

Figure 5-3 Sysdump screen with no SD card inserted

To dump sysdump to PC:

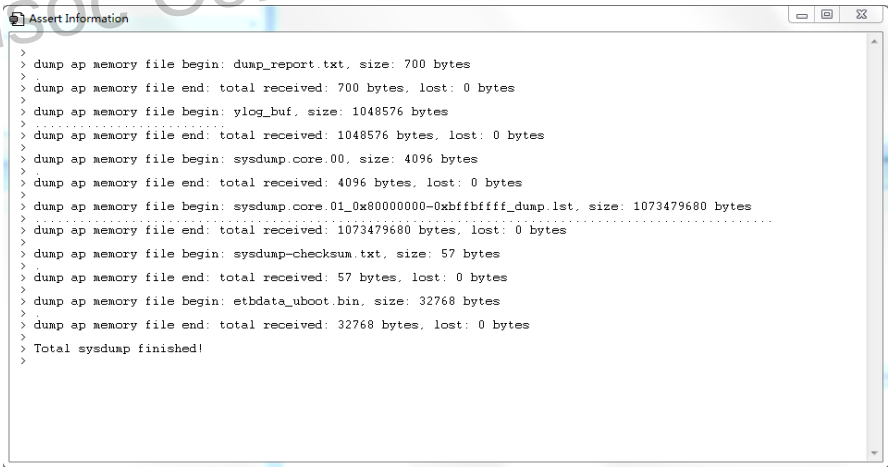
a. Open Logel, as shown in **Figure 5-4**, click the **Option** > **Tool configure** > **On/Off**, select **Auto Identify AP Dump Pt** and then close Logel.

Figure 5-4 Auto Identify AP Dump Pt window



b. Open Logel again, connect the DUT with the PC through the USB cable, click the **Capture** button and export the Dump file to the PC as shown in **Figure 5-5**.

Figure 5-5 Dump to PC window



Unisoc Confidential For longcheer

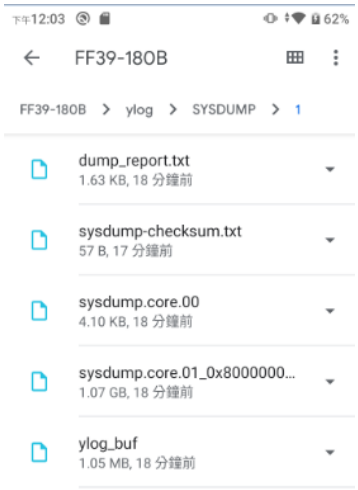
----End

5.1.3 Log storage

The sysdump log is stored in the **ylog /SYSDUMP** path. To quickly check the integrity of Sysdump log, you can first check if the size of any log file is zero, and then if there is any log file missing.

- The files generated in Sysdump include the dump_report.txt file, which contains **Mem_num is xxx** in the content.
- The name of the log files generated in sysdump are all in the format of **sysdump.core.xx_0x..**, as shown in **Figure 5-6**.

Figure 5-6 Sysdump log list



For example, **Mem_num is 10**, the sysdump file name should be from sysdump.core.00*xxx* ~sysdump.core.10*xxx*. If the number of logs in sysdump.core.*xx* format does not match Mem_num, it indicates that sysdump log generation is incomplete.

Unisoc Confidential For longcheer

5.2 Memory log of assert problem

5.2.1 YLog settings

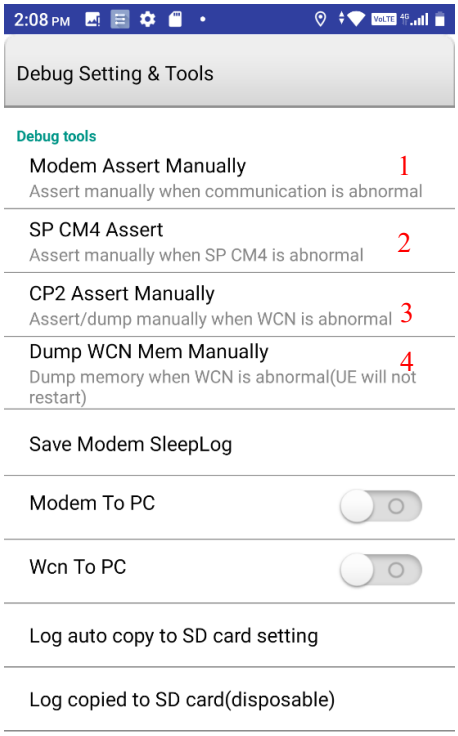
- In Userdebug version, **Modem Reset** and **CP2 Reset** are disabled by default.
- In User version, **Modem Reset** and **CP2 Reset** are enabled by default. You need to disable them in **YLog** > **settings** >**Log setting** >**modem log setting /Connectivity Log setting** in Engineer Mode before testing.

5.2.2 Log capture

Memory dump is automatically captured in case of ASSERT, and there is prompt on screen.

Assert can be manually triggered in **YLog** > **settings** > **Debug Setting & Tools** in case of system exception, as shown in **Figure 5-7**.

Figure 5-7 Debug setting & tools



Unisoc Confidential For longcheer

1. **Modem Assert Manually:** it is the option to assert Modem (CP0) manually. Tapping the option pops up a prompt box and you need to follow the on-screen instructions for correct operation. Manual Modem assert generates the .dmp and minidump files of CP0.
2. **SP CM4 Assert:** it is used for watchdog debugging, which triggers CM4 assert.
3. **CP2 Assert Manually**: it is the option to assert CP2 manually. Clicking the option pops up a prompt box and you need to follow the instructions to generate wcn .dmp.

   Assert requires AP to interact with CP2 that is, AP first sends the Assert command to CP2 and CP2 actively asserts after receiving the command and informs AP. AP starts dumping after receiving the Assert status.
4. **Dump WCN Mem Manually:** it is the option to assert WCN manually. Clicking the option pops up a prompt box and you need to follow the on-screen instructions to generate wcn .dmp.

   CP2 cannot process the Assert command for some reason after the **CP2 Assert Manually** option is clicked, which leads to the failure in processing all subsequent operations. In such situation, you can select **Dump WCN Mem Manually**

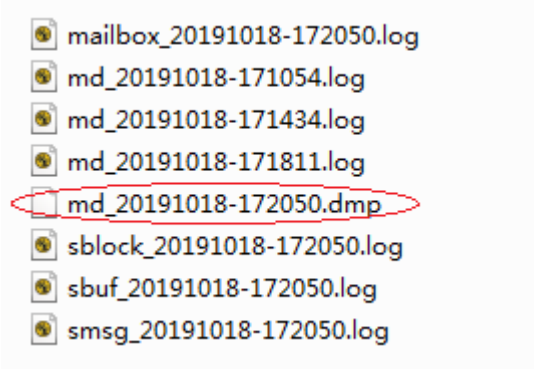
NOTE

It is suggested to use the Assert command first to ensure that WCN dump file is saved to memory successfully. You can tap **Dump WCN Mem Manually** again to save dump files if the system does not assert normally.

5.2.3 Log storage

A file with the suffix .dmp is generated after the dump process ends, as shown in **Figure 5-8**.

Figure 5-8 Assert mem log



5.3 Audio log

Capture audio logs to analyze call quality issues.

5.3.1 YLog settings

Set YLog to Audio Scene.

5.3.2 Log capture

Establish a voice call or a video call, and capture the logs on both MO and MT sides during the call.

Unisoc Confidential For longcheer

5.3.3 Log storage

- For chips with an independent AGDSP, an Audio folder and ag.log are generated. An Audio mem log is generated after modem log replay.
- For chips without an independent AGDSP, an audio mem log is generated after modem log replay.

5.3.4 Log replay

Step 1 Use Logel to play back the log in the **YLog/modem log** directory to generate an Audio mem log file with the suffix iq.mem, as shown in **Figure 5-9**.

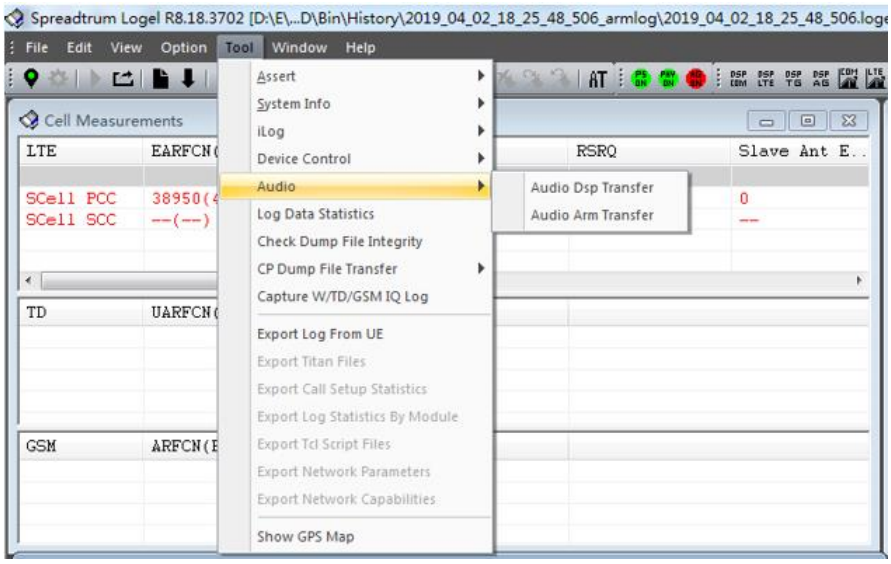
Figure 5-9 Audio mem log

名称	修改日期	类型	大小
2019_04_02_18_25_48_506	2019/4/2 18:26	文件夹	
2019_04_02_18_25_48_506.cap	2019/4/2 18:26	CAP 文件	211 KB
2019_04_02_18_25_48_506.ip	2019/4/2 18:25	IP 文件	0 KB
2019_04_02_18_25_48_506.iq	2019/4/2 18:25	IQ 文件	0 KB
2019_04_02_18_25_48_506.logel	2019/4/2 18:26	LOGEL 文件	172,018 KB
2019_04_02_18_25_48_506.lst	2019/4/2 18:26	LST 文件	1 KB
2019_04_02_18_25_48_506.wrrc_log	2019/4/2 18:25	WRRC_LOG 文件	0 KB
2019_04_02_18_25_48_506.wvoice	2019/4/2 18:25	WVOICE 文件	0 KB
2019_04_02_18_25_48_506.xdsp_log	2019/4/2 18:25	XDSP_LOG 文件	0 KB
2019_04_02_18_25_48_506_bt.cap	2019/4/2 18:25	CAP 文件	0 KB
2019_04_02_18_25_48_506_capture_only_wcn	2019/4/2 18:25	LOG 文件	0 KB
2019_04_02_18_25_48_506_dsp.dump	2019/4/2 18:25	DUMP 文件	0 KB
2019_04_02_18_25_48_506_dsp.org	2019/4/2 18:25	ORG 文件	0 KB
2019_04_02_18_25_48_506_dsp_ag.dump	2019/4/2 18:25	DUMP 文件	0 KB
2019_04_02_18_25_48_506_dsp_ag.org	2019/4/2 18:25	ORG 文件	0 KB
2019_04_02_18_25_48_506_dsp_ag_audio.mem	2019/4/2 18:25	MEM 文件	0 KB
2019_04_02_18_25_48_506_dsp_ag_iq.mem	2019/4/2 18:25	MEM 文件	0 KB
2019_04_02_18_25_48_506_dsp_audio.mem	2019/4/2 18:25	MEM 文件	0 KB
2019_04_02_18_25_48_506_dsp_iq.mem	2019/4/2 18:25	MEM 文件	0 KB
2019_04_02_18_25_48_506_dsp_lte.dump	2019/4/2 18:25	DUMP 文件	0 KB
2019_04_02_18_25_48_506_dsp_lte.org	2019/4/2 18:25	ORG 文件	138,724 KB
2019_04_02_18_25_48_506_dsp_lte_audio.mem	2019/4/2 18:25	MEM 文件	0 KB
2019_04_02_18_25_48_506_dsp_lte_iq.mem	2019/4/2 18:25	MEM 文件	0 KB
2019_04_02_18_25_48_506_dsp_tg.dump	2019/4/2 18:25	DUMP 文件	0 KB
2019_04_02_18_25_48_506_dsp_tg.org	2019/4/2 18:25	ORG 文件	3,330 KB
2019_04_02_18_25_48_506_dsp_tg_audio.mem	2019/4/2 18:25	MEM 文件	0 KB
2019_04_02_18_25_48_506_dsp_tg_iq.mem	2019/4/2 18:25	MEM 文件	3,362 KB
2019_04_02_18_25_48_506_gprs.cap	2019/4/2 18:25	CAP 文件	1 KB
2019_04_02_18_25_48_506_lte.cap	2019/4/2 18:25	CAP 文件	1 KB
2019_04_02_18_25_48_506_modem	2019/4/2 18:25	配置设置	1 KB
2019_04_02_18_25_48_506_mux.cap	2019/4/2 18:25	CAP 文件	1 KB
2019_04_02_18_25_48_506_trace	2019/4/2 18:25	文本文档	0 KB
2019_04_02_18_25_48_506_vt_down.bin	2019/4/2 18:25	BIN 文件	0 KB
2019_04_02_18_25_48_506_vt_up.bin	2019/4/2 18:25	BIN 文件	0 KB
2019_04_02_18_25_48_506_w.iq	2019/4/2 18:25	IQ 文件	0 KB
2019_04_02_18_25_48_506_wcn.dump	2019/4/2 18:25	DUMP 文件	0 KB
2019_04_02_18_25_48_506_wcn_dsp.org	2019/4/2 18:25	ORG 文件	0 KB

Unisoc Confidential For longcheer

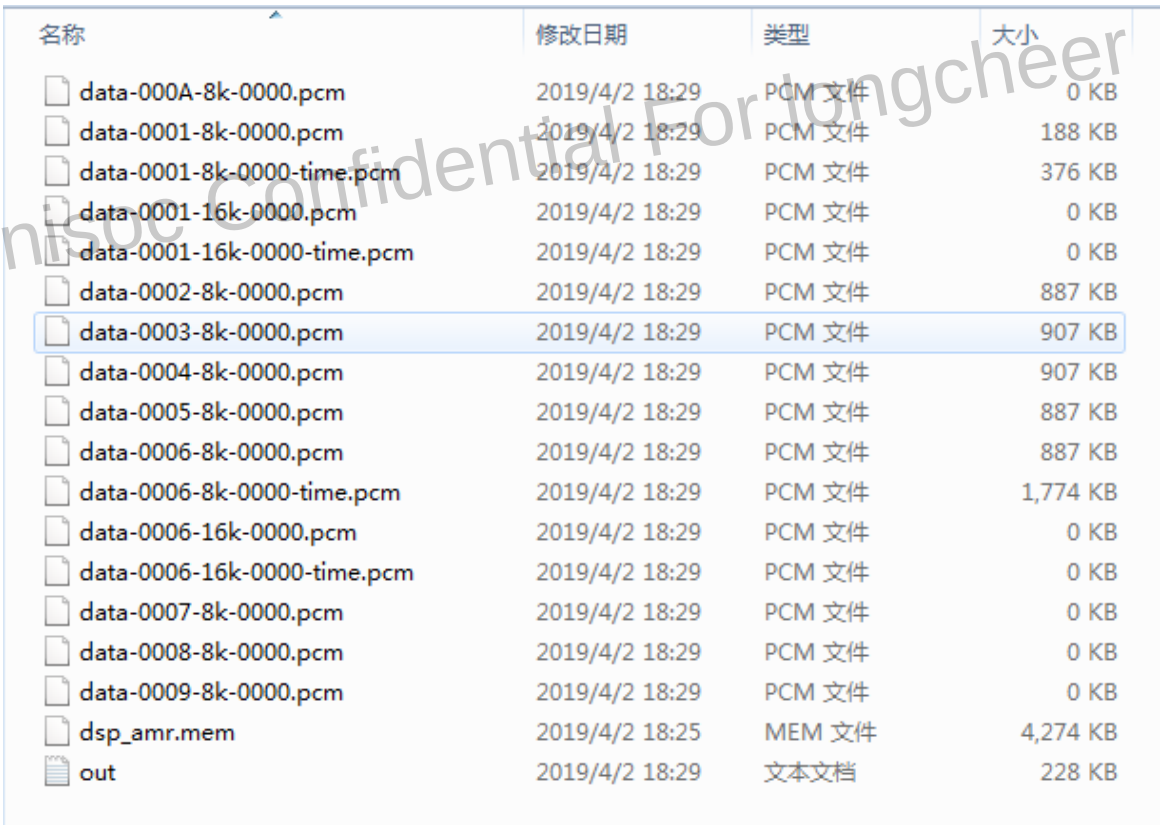
Step 2 Click **Tool** > **Audio** > **Audio dsp transfer** in Logel, as shown in **Figure 5-10**.

Figure 5-10 Audio dsp transfer



Load iq.mem to convert the captured dsp pcm data as shown in **Figure 5-11**.

Figure 5-11 DSP PCM data



名称	修改日期	类型	大小
data-000A-8k-0000.pcm	2019/4/2 18:29	PCM 文件	0 KB
data-0001-8k-0000.pcm	2019/4/2 18:29	PCM 文件	188 KB
data-0001-8k-0000-time.pcm	2019/4/2 18:29	PCM 文件	376 KB
data-0001-16k-0000.pcm	2019/4/2 18:29	PCM 文件	0 KB
data-0001-16k-0000-time.pcm	2019/4/2 18:29	PCM 文件	0 KB
data-0002-8k-0000.pcm	2019/4/2 18:29	PCM 文件	887 KB
data-0003-8k-0000.pcm	2019/4/2 18:29	PCM 文件	907 KB
data-0004-8k-0000.pcm	2019/4/2 18:29	PCM 文件	907 KB
data-0005-8k-0000.pcm	2019/4/2 18:29	PCM 文件	887 KB
data-0006-8k-0000.pcm	2019/4/2 18:29	PCM 文件	887 KB
data-0006-8k-0000-time.pcm	2019/4/2 18:29	PCM 文件	1,774 KB
data-0006-16k-0000.pcm	2019/4/2 18:29	PCM 文件	0 KB
data-0006-16k-0000-time.pcm	2019/4/2 18:29	PCM 文件	0 KB
data-0007-8k-0000.pcm	2019/4/2 18:29	PCM 文件	0 KB
data-0008-8k-0000.pcm	2019/4/2 18:29	PCM 文件	0 KB
data-0009-8k-0000.pcm	2019/4/2 18:29	PCM 文件	0 KB
dsp_amr.mem	2019/4/2 18:25	MEM 文件	4,274 KB
out	2019/4/2 18:29	文本文档	228 KB

Unisoc Confidential For longcheer

Step 3 Click **Tool** > **Audio** > **Audio arm transfer** in Logel, then click **OK** to convert the captured ARM PCM data, as shown in Figure 5-12.

Figure 5-12 ARM PCM data

----End

5.4 Power consumption problem log

5.4.1 YLog settings

Close all logs before power consumption test and enable user scene in retest (only open AP log and close Modem log).

5.4.2 Log export

You can export log using Log4Android2PC or ADB pull command.

Unisoc Confidential For longcheer

5.5 MonkeyTest log

5.5.1 YLog settings

Confirm the settings below before conducting Monekeytest:

- Enable **AP Log** and close **Modem Log**.
- Enable the **SysdumpEnable** function in **YLog > Settings >SysdumpEnable** to ensure Sysdump Log can be captured.
- Disable **Modem Reset** and **CP2 Reset** in **YLog** > **Settings** > **Log setting** > **Modem log setting/Connectivity Log Setting**.

5.5.2 Log export

You can export log using Log4Android2PC or ADB pull command.

It is recommended to export logs once a day and clear the log history for long-running (such as 7x24h) test.

6 Points for attention

1. In User version, the YLog switch and the Sysdump Enable switch are disabled by default, which needs to be enabled before testing. The Modem Assert reset and WCN Assert reset switches should be disabled.
2. If the Modem To PC function is enabled, need to capture Modem Log using the online tool Logel.
3. Use Log4Android2PC to export the captured logs. Ensure an SD card of 16GB Class 10 and above are inserted to enable the auto log copy to SD card function.
4. Provide corresponding AP Log and Modem Log for common issues, whereas complete mem file for special scene issues such as modem assert, wcn assert, sysdump, audio mem, and others.
5. Confirm log effectiveness based on log size and time point of the issue. In another word, the log size is within a reasonable range and the cut-off time of the log covers the time point of the issue.
6. The data of the five buffers (main, system, radio, vents, crash) in the current Android Log are written to the same file. You need to install Python tool on PC to run the analyzer.py script to extract them.
7. YLog is designed not to delete non-YLog module generated content and corefile is not generated by YLog so the delete button on YLog screen cannot delete corefile.

Unisoc Confidential For longcheer

7 Frequently asked questions

1. Q: Trigger sysdump immediately after changing the sysdump function setting and the state is not saved after restart.

   A: This is not an issue, because property information has not been synchronized to disk if sysdump is triggered immediately after the setting change.

2. Q: SD card is formatted as an internal card, which cannot conduct sysdump.

   A: This is not an issue, because the file system does not support internal card. The SD card must be formatted as an external card.

3. Q: The sysdump process is terminated and the screen displays "init_mmc_fat failed, please check SD Card!" Press volume up, volume down, or home to exit.

   A: This is because the SD card initialization is failed or the SD card file format is not supported by the system. Sysdump only supports exFAT and FAT32. In this case, you need to check if the SD card is inserted, or replace the SD card and format it as an external card.

4. Q: The sysdump process is terminated and the screen displays "SD Card is damaged", the screen prompts: press volume down to format the SD card and then continue the sysdump process, or press volume up to terminate the sysdump process.

   A: This is due to mmc initialization error init_mmc_fat which can be found in the Uboot log. In this case format or replace the SD card.

5. Q: The sysdump process is terminated and the screen gets stuck in a file.

   A: The sysdump files are relatively large, so please be patient. If the scene is very long, you can find do_fat_write in Uboot log which indicates exception in calling FAT inteface. In this case, format or replace the SD card.

6. Q: The sysdump process is terminated and the screen displays Reset mode: manual dump stuck without any other information.

   A: In the Uboot log you can find that the file system check is already supported, but the fat_checksys function call result of the FS check space is abnormal. In this case, format or replace the SD card.

7. Q: It restarts directly without entering sysdump process upon exception in Userdebug version.

   A: Check the serial port or kernel log, you can find that the exception has occurred in kernel in the early stage of booting (such as "[0.533670]"), where the sysdump function has not been enabled.

UNISOC Confidential For longcheer